<Team 1>

Recasting MAPF Problems as Group Actions of Automorphism Groups Generated by Graph Edge Transpositions

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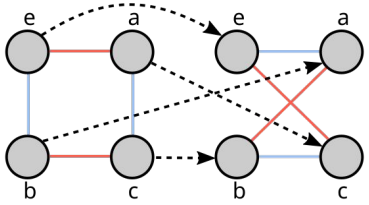
Motivation & Problem Statement



$\phi(e) = e$
$\phi(a) = c$
$\phi(b) = a$
$\phi(c) = b$
$\phi = (acb)$

- **Problem**: MAPF as graph traversal problem
- **Alternative**: MAPF as **constrained word problem**
- **How**: Recasting MAPF Problems as **Group Actions** of **Automorphism Groups Generated by Graph Edge Transpositions**
- **Preliminaries**: Graph $G = (V, E)$, Agents $A = \{1, \ldots, n\}$, Starts S, Goals G
- **Group Action** ($H \curvearrowright V$): A function that maps a group H and a set V to V itself
- **Automorphism Group** ($\mathrm{Aut}(V)$): All bijective functions (permutations) from V to V
- **Generated by Graph Edge Transpositions** ($H = \langle (u\ v) \mid (u, v) \in E \rangle$): Composition of graph edges (adjacent transpositions) produce all members of group H
- **Constrained Word Problem**: Finding a constrained sequence of generator elements in H that compose together to generate the S to G permutation

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Proposed Solution & Key Ideas

- **Importance**: Exploit symmetries and invariants of group structure
- **Research Gap**: Existing literature does not formally treat MAPF as group-theoretic problems, but has been implicitly borrowing ideas from it for various applications
- **Key Ideas**:
  - **Anonymous MAPF**: Searches in the quotient group of the symmetric group under agent relabeling symmetry
  - **PBS**: Implicitly stabilizes paths higher priority agents take. Lower priority agents plan paths in stabilizer chains generated by Schreier–Sims
  - **Map Partitioning**: Decompose map into Young subgroup via min-cuts
  - **Group Structure**: Natural structure for symmetries, invariants, permutations, automorphisms, etc